

KARTHICK T

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English, Tamil, Kannada, Hindi, Telugu

PROFESSIONAL SUMMARY

Aspiring Robotics and Automation Engineer passionate about developing intelligent robotic systems, industrial automation, and autonomous solutions. Committed to applying technical expertise, problem-solving, and innovation to design, optimize, and enhance robotic technologies while contributing to organizational growth.

TECHNICAL SKILLS

Area Of Interest

Autonomous Mobile Robot
ROS2 Development
Industrial Automation

Programming

Python, Java
C Programming
ROS2, PLC (LL)

Design & Analysis

SolidWorks, Creo, Fusion 360
RoboDK, RViz, Gazebo
Proteus, Tinkercad, Wokwi

INTERNSHIP EXPERIENCE

Hanon Systems - Manufacturing Sector

Assisted in the manufacturing and assembly of HVAC systems, compressors, and condenser units while gaining exposure to ABB and Yaskawa industrial robots.

Steel Authority of India Limited (SAIL) - Manufacturing Sector

Studied hot and cold rolling mill operations, material handling, quality inspection, and end-to-end steel manufacturing workflows in an integrated production and automation environment.

PROJECT EXPERIENCE

ROS 2 Autonomous Mobile Robot

Developed a four-wheel Autonomous Mobile Robot (AMR) using ROS 2 Jazzy and Gazebo Harmonic with LiDAR-based SLAM, Jump Point Search (JPS), and Regulated Pure Pursuit (RPP). Validated autonomous navigation using ROS 2 Control and Navigation2.

Vision-Guided Agricultural Fruit Harvesting Manipulator

Developed a vision-guided agricultural fruit harvesting system integrating a multi-joint robotic manipulator, custom blade-integrated gripper, and YOLO-based fruit detection. Implemented Raspberry Pi 5-ESP32 control for automated fruit localization, harvesting, and safe collection.

Voice-Controlled Vision-Based Pick-and-Place Robotic Manipulator

Developed an intelligent voice-controlled vision-based robotic manipulator that combines Whisper speech recognition, YOLOv8n object detection, RGB-D depth sensing, 3D object localization, inverse kinematics, and ESP32-based servo control to autonomously identify and pick user-specified objects.

Solenoid Valve-Based Autonomous Four-Wheel Steering Mobile Robot

Designed and developed a solenoid valve-based Autonomous Mobile Robot (AMR) with switchable two-wheel and four-wheel steering modes, enabling obstacle avoidance, autonomous navigation, and enhanced maneuverability.

ACHIEVEMENTS

Smart Home Automation – SIH 2023

Selected for Round 2 of Smart India Hackathon with an Autonomous Smart Environment solution.

Third Place – Technical Paper Presentation

Topic: Robotic Arm | Chennai Institute of Technology (CIT).

Robotic Gripper Design Pattern

Patent Filed – Complex Material Handling Robotic Gripper
Patent Filed – Tri-Claw Knuckle Lift Gripper System

EDUCATION & LEADERSHIP

B.E - Robotics And Automation

Anna University, MIT Campus.

Web Developer Team Head - MITRA Club

Anna University, MIT Campus.

Head Of The Team – SIH 2023

Smart India Hackathon, MIT Campus.

Head Of The Team A – SAIL Intern

Manufacturing Sector, Steel Authority of India Limited.